

Community Library Membership & Book Lending System

Stored Procedures I have used in code:

-- calculate_member_fines

```
CREATE OR REPLACE FUNCTION calculate_member_fine(p_member_id INT)
```

```
RETURNS DECIMAL AS $$
```

```
DECLARE
```

```
    v_total_fine DECIMAL := 0;
```

```
BEGIN
```

```
    SELECT COALESCE(SUM("FineAmount"), 0) INTO v_total_fine
```

```
    FROM fines
```

```
    WHERE "MemberId" = p_member_id
```

```
    AND "PaymentStatus" = 'Pending'
```

```
    AND "IsActive" = true;
```

```
    RETURN v_total_fine;
```

```
END;
```

```
$$ LANGUAGE plpgsql;
```

-- get_member_borrowing_summary

```
CREATE OR REPLACE FUNCTION get_member_borrowing_summary(p_member_id INT)
```

```
RETURNS TABLE(
```

```
    active_borrowings INT,
```

```
    returned_borrowings INT,  
    overdue_borrowings INT,  
    total_fine DECIMAL  
) AS $$  
  
DECLARE  
  
    v_active_count INT := 0;  
    v_returned_count INT := 0;  
    v_overdue_count INT := 0;  
    v_total_fine DECIMAL := 0;  
  
BEGIN  
  
    -- Count active borrowings (currently borrowed)  
    SELECT COUNT(*) INTO v_active_count  
    FROM borrowings  
    WHERE member_id = p_member_id  
    AND status = 'Borrowed'  
    AND is_active = true;  
  
    -- Count returned borrowings  
    SELECT COUNT(*) INTO v_returned_count  
    FROM borrowings  
    WHERE member_id = p_member_id  
    AND status = 'Returned'  
    AND is_active = true;  
  
    -- Count overdue borrowings  
    SELECT COUNT(*) INTO v_overdue_count  
    FROM borrowings
```

```

WHERE member_id = p_member_id
AND status = 'Borrowed'
AND due_date < NOW()
AND is_active = true;

-- Calculate total unpaid fine
SELECT COALESCE(SUM(fine_amount), 0) INTO v_total_fine
FROM fines
WHERE member_id = p_member_id
AND payment_status = 'Pending'
AND is_active = true;

RETURN QUERY
SELECT v_active_count, v_returned_count, v_overdue_count, v_total_fine;
END;
$$ LANGUAGE plpgsql;

```

Entities:

Admin Entity:

```

using System;

namespace LibrarySystem.DataAccess.Entities
{
    public class Admin{
        public int Id {get; set;}

        public string Name {get; set;} = string.Empty;

        public string PhoneNum {get; set;} = string.Empty;
    }
}

```

```

    public string Password {get; set;} = string.Empty;
    public bool IsActive {get; set;}
    public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
    public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;
}
}

```

MemberEntity

```

using System;
using LibrarySystem.DataAccess.Enums;
namespace LibrarySystem.DataAccess.Entities
{
    public class Member{
        public int Id {get; set;}
        public string Name {get; set;} = string.Empty;
        public string Email {get; set;} = string.Empty;
        public string PhoneNum {get; set;} = string.Empty;
        public string Password {get; set;} = string.Empty;
        public MembershipType MembershipType {get; set;} = MembershipType.Basic;
        public MembershipStatus MembershipStatus {get; set;} = MembershipStatus.Active;
        public bool IsActive {get; set;} = true;
        public int AllowedBorrowingCount {get; set;}
        public int CurrentBorrowedCount {get; set;}
        public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
        public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;
    }
}

```

```
//navigation(foreign keys)

public virtual ICollection<Borrowing> Borrowings {get; set;} = new List<Borrowing>();

public virtual ICollection<Fine> Fines {get; set;} = new List<Fine>();

}

}
```

BookCategoryEntity

```
using System;

namespace LibrarySystem.DataAccess.Entities
{
    public class BookCategory{
        public int Id {get; set;}

        public string CategoryName {get;set;} = string.Empty;

        public bool IsActive {get; set;} = true;

        public DateTime CreatedAt {get; set;} = DateTime.UtcNow;

        public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;


        public virtual ICollection<Book> Books {get; set;} = new List<Book>();

    }

}
```

BookCopyEntity

```
using System;

namespace LibrarySystem.DataAccess.Entities
{
    public class BookCopy{
        public int Id { get; set;}
        public int BookId {get; set;}
        public int BookCopyId {get; set;}
        public bool IsAvailable {get; set;} = true;
        public bool IsBorrowed {get; set;} = false;
        public bool IsDamaged {get; set;} = false;
        public string ConditionNotes {get; set;} = string.Empty;
        public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
        public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;
        public virtual Book Book {get; set;} = null!;
        public virtual ICollection<Borrowing> Borrowings {get; set;} = new List<Borrowing>();
    }
}
```

BookEntity

```
using System;

namespace LibrarySystem.DataAccess.Entities
{
```

```

public class Book{
    public int Id {get; set;}
    public int CategoryId {get; set;}
    public string Title {get; set;} = string.Empty;
    public string Author {get; set;} = string.Empty;
    public string ISBN {get; set;} = string.Empty;
    public int PublicationYear {get; set;}
    public int NoOfCopies {get; set;}
    public bool IsActive {get; set;} = true;
    public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
    public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;

    public virtual BookCategory Category {get; set;} = null!;
    public virtual ICollection<BookCopy> BookCopies {get; set;} = new List<BookCopy>();
    public virtual ICollection<Borrowing> Borrowings {get; set;} = new List<Borrowing>();
}
}

```

BorrowingRulesEntity

```

using System;
using LibrarySystem.DataAccess.Enums;
namespace LibrarySystem.DataAccess.Entities
{
    public class BorrowingRules{
        public int Id {get; set;}

```

```

    public MembershipType MembershipType {get; set;} = MembershipType.Basic;
    public int MaxActiveBorrowings {get; set;}
    public int MaxBorrowDays {get; set;}
    public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
    public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;
}
}

```

BookBorrowEntity

```

using System;
using LibrarySystem.DataAccess.Enums;
namespace LibrarySystem.DataAccess.Entities
{
    public class Borrowing{
        public int Id { get; set;}
        public int MemberId { get; set;}
        public int BookCopyId { get; set;}
        public int BookId {get; set;}
        public DateTime BorrowedDate {get; set;} = DateTime.UtcNow;
        public DateTime DueDate {get; set;} = DateTime.UtcNow;
        public BookBorrowStatus Status {get; set;} = BookBorrowStatus.Borrowed;
        public DateTime? MemberReturnedDate {get; set;}
        public decimal FineAmount {get; set;} = 0;
        public bool IsActive {get; set;} = true;
        public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
        public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;
    }
}

```



```

    public virtual Member Member {get; set;} = null!;
    public virtual BookCopy BookCopy {get; set;} = null!;
    public virtual Book Book {get; set;} = null!;
    public virtual ICollection<Fine> Fines {get; set;} = new List<Fine>();
}
}

```

FineEntity

```

using System;
using LibrarySystem.DataAccess.Enums;
namespace LibrarySystem.DataAccess.Entities
{
    public class Fine{
        public int Id {get; set;}
        public int BorrowingId {get; set;}
        public int MemberId {get; set;}
        public decimal FineAmount {get; set;}
        public string FineReason {get; set;} = string.Empty;
        public FinePaymentStatus PaymentStatus {get; set;} = FinePaymentStatus.Pending;
        public DateTime? PaymentDate {get; set;}
        public bool IsActive { get; set; } = true;
        public DateTime CreatedAt {get; set;} = DateTime.UtcNow;
        public DateTime UpdatedAt {get; set;} = DateTime.UtcNow;
        public virtual Member Member {get; set;} = null!;
        public virtual Borrowing Borrowing {get; set;} = null!;}}

```

Enums:

BookBorrowStatus

```
using System;

namespace LibrarySystem.DataAccess.Enums
{
    public enum BookBorrowStatus
    {
        Borrowed = 1,
        Returned = 2,
        Overdue = 3
    }
}
```

FinePaymentStatus

```
using System;

namespace LibrarySystem.DataAccess.Enums
{
    public enum FinePaymentStatus
    {
        Paid = 1,
        Pending = 2
    }
}
```

MembershipStatus

namespace LibrarySystem.DataAccess.Enums

```
{  
    public enum MembershipStatus  
    {  
        Active = 1,  
        Inactive = 2  
    }  
}
```

MembershipType

namespace LibrarySystem.DataAccess.Enums

```
{  
    public enum MembershipType  
    {  
        Basic = 1,  
        Premium = 2,  
        Student = 3  
    }  
}
```